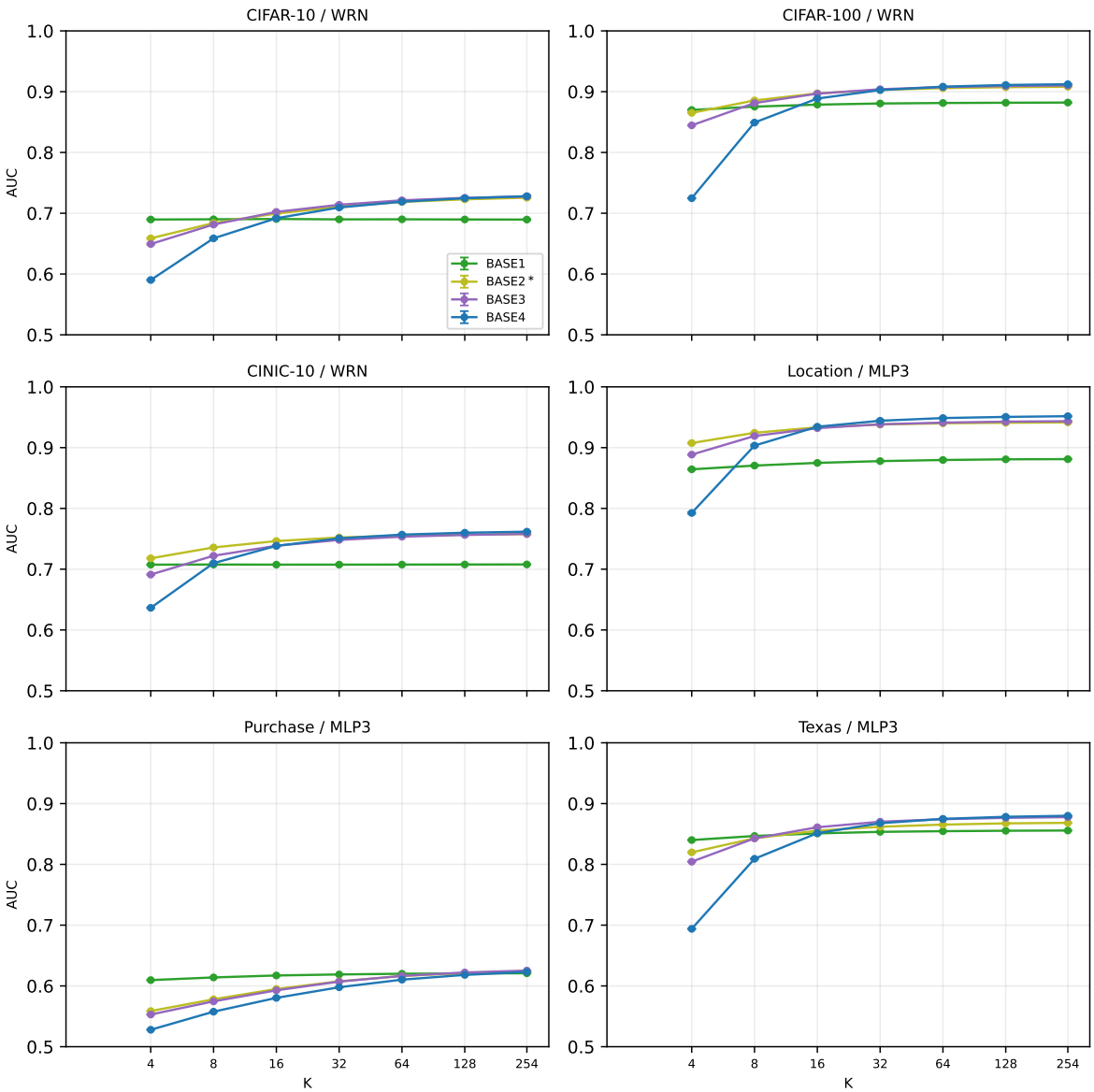

Figure 1: AUC vs. shadow-model budget $K \in \{4, 8, 16, 32, 64, 128, 254\}$, WRN / MLP3 testbeds. BASE1 = pooled-exponential model on the loss; BASE3 = class-separated means with a per-sample pooled variance; BASE4 = full class-conditional Gaussian (equivalent to LiRA online); BASE2* = BASE3 with the per-sample variance fixed to $\sigma \equiv 1$, isolating class-mean separation, with score $\mathrm{BASE2}_i^* = (\hat{\mu}_{i,1} - \hat{\mu}_{i,0})\,(z_i - \frac{1}{2}(\hat{\mu}_{i,1} + \hat{\mu}_{i,0}))$. Mean ± 1 SE over 32 leave-one-out replicates per configuration (every shadow model audited as the held-out target), antithetic-paired design (128 pairs, $K = 254$ max), online setting only. ROC curves throughout this report are interpolated onto a common log-spaced FPR grid and aggregated as the mean.

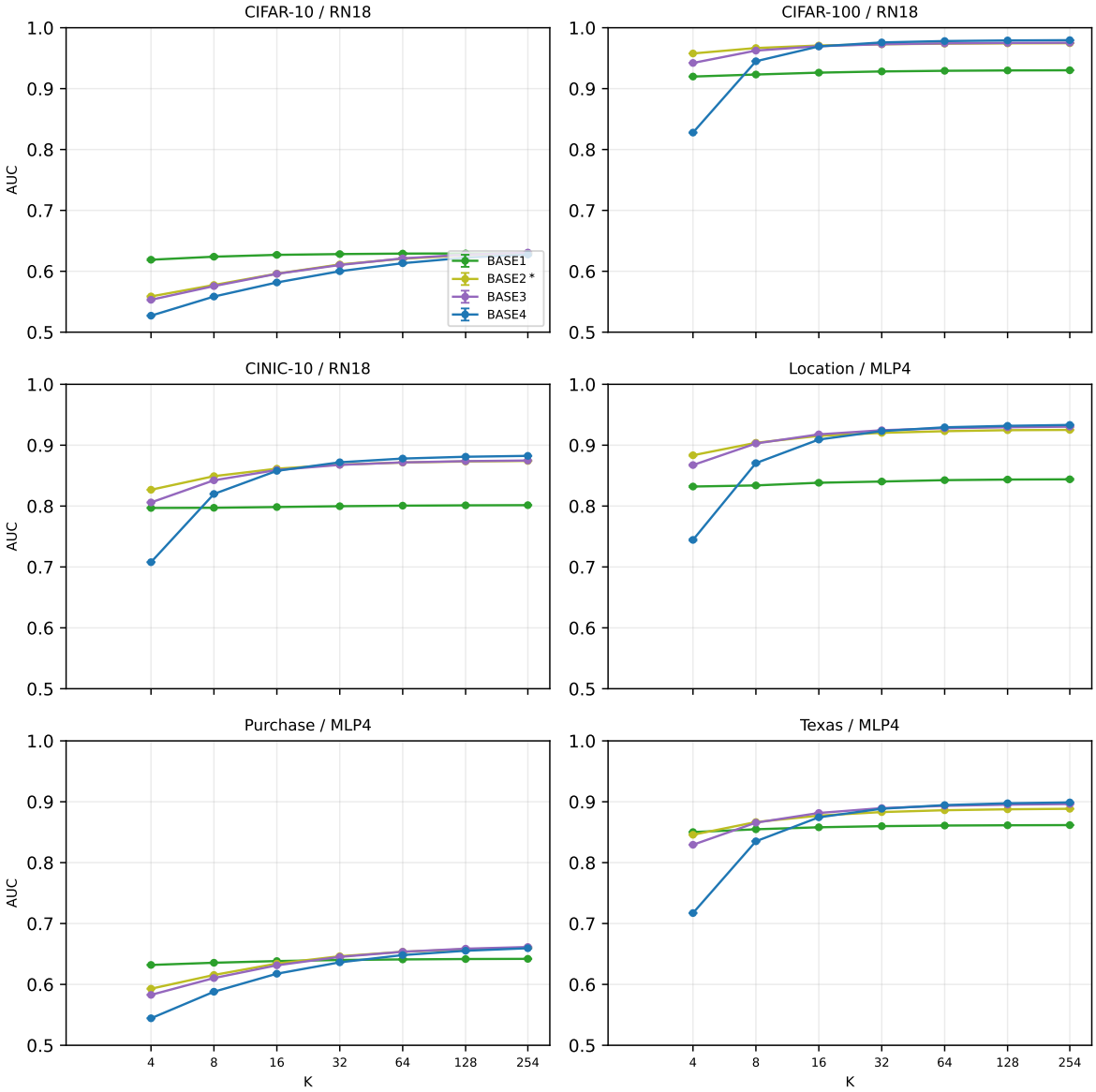

Figure 2: AUC vs. K, RN18 / MLP4 testbeds. BASE1, BASE2*, BASE3, BASE4 (online). Mean ± 1 SE over 32 replicates.

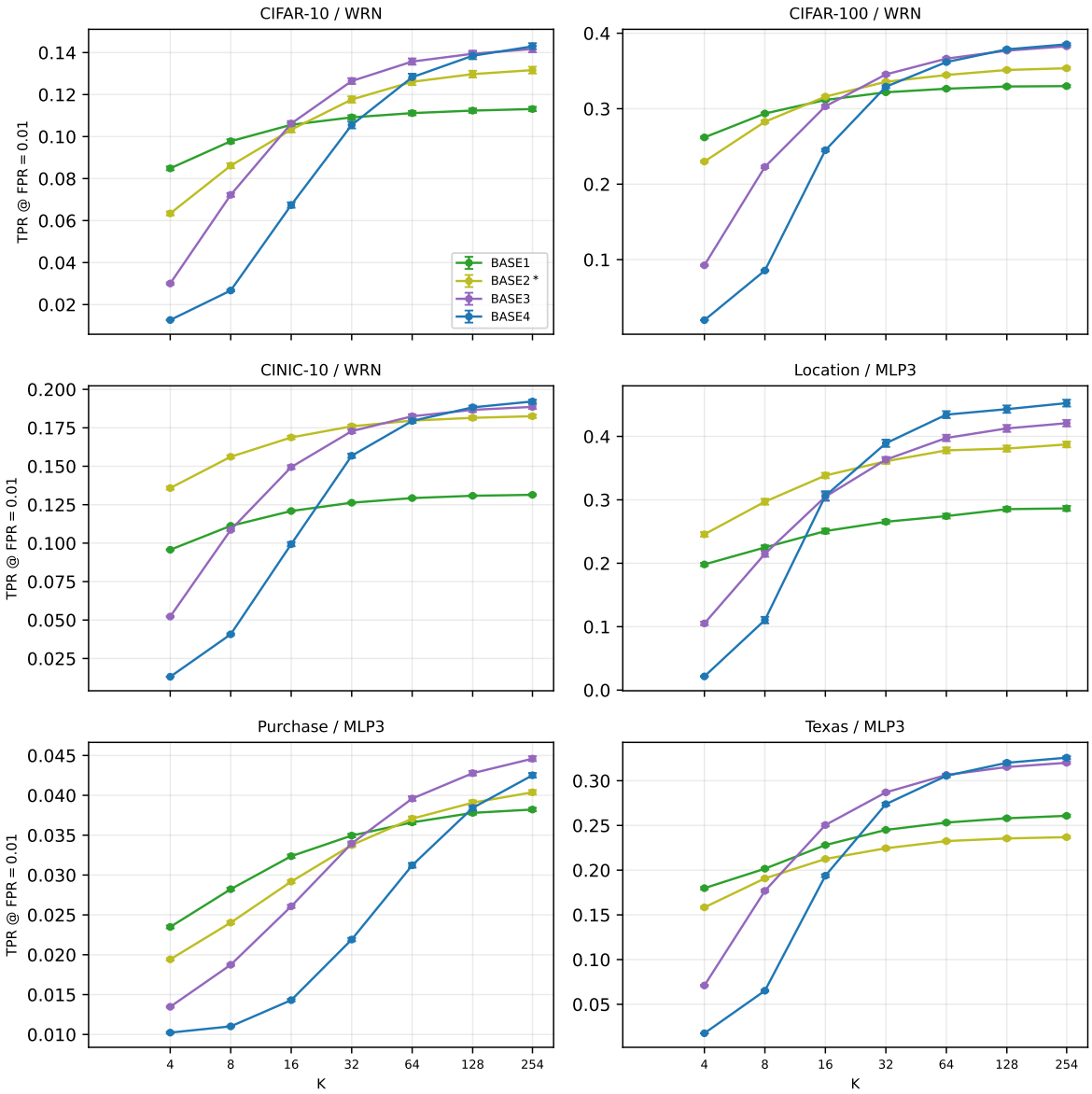

Figure 3: TPR at FPR= 0.01 vs. K, WRN / MLP3 testbeds. BASE1, BASE2*, BASE3, BASE4. Mean ± 1 SE over 32 replicates. Note BASE4's collapse at $K \leq 8$, the regime where simpler BASE1 / BASE2* dominate, consistent with per-class variance estimation being the bottleneck at small K.

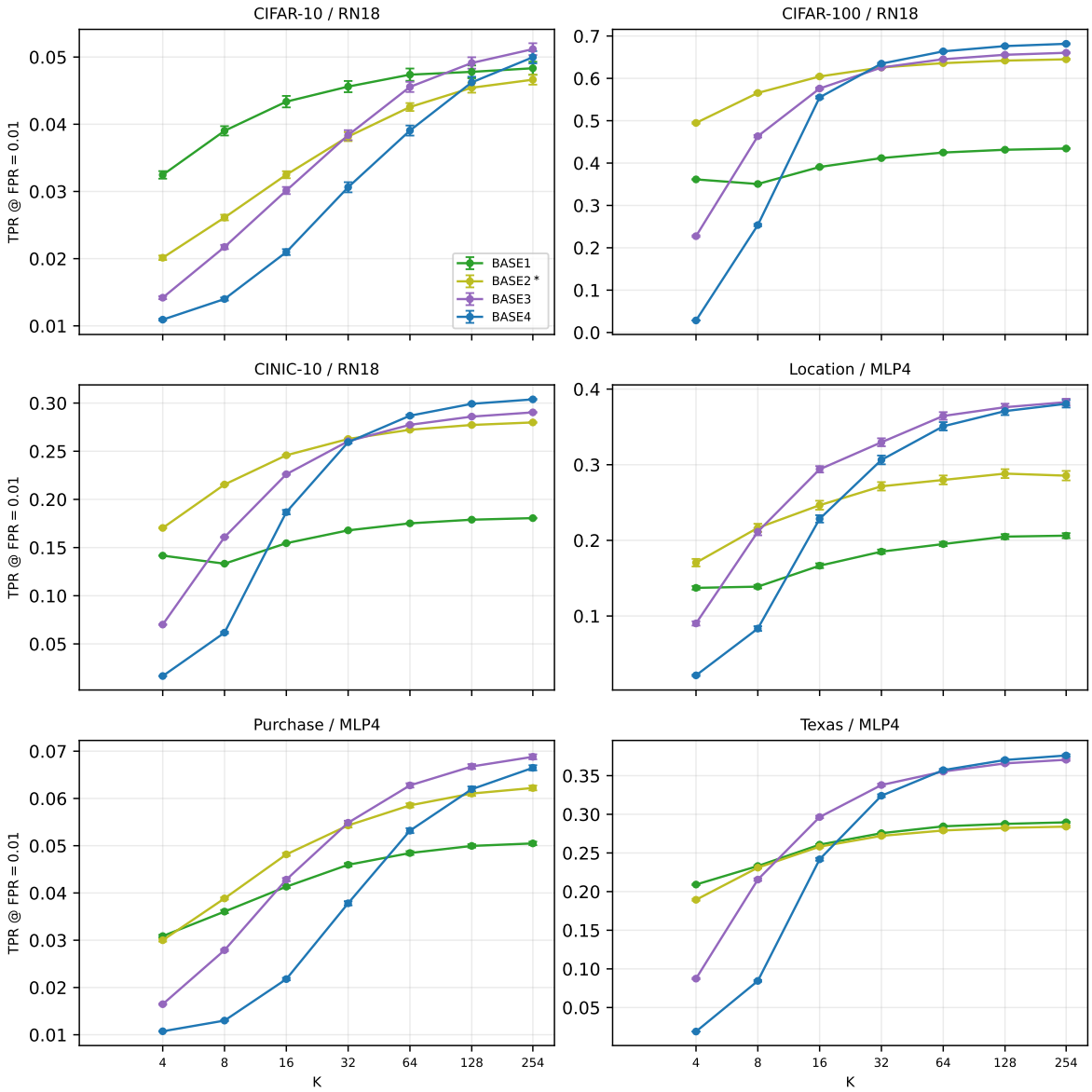

Figure 4: TPR at FPR= 0.01 vs. K, RN18 / MLP4 testbeds. BASE1, BASE2*, BASE3, BASE4. Mean ± 1 SE over 32 replicates.

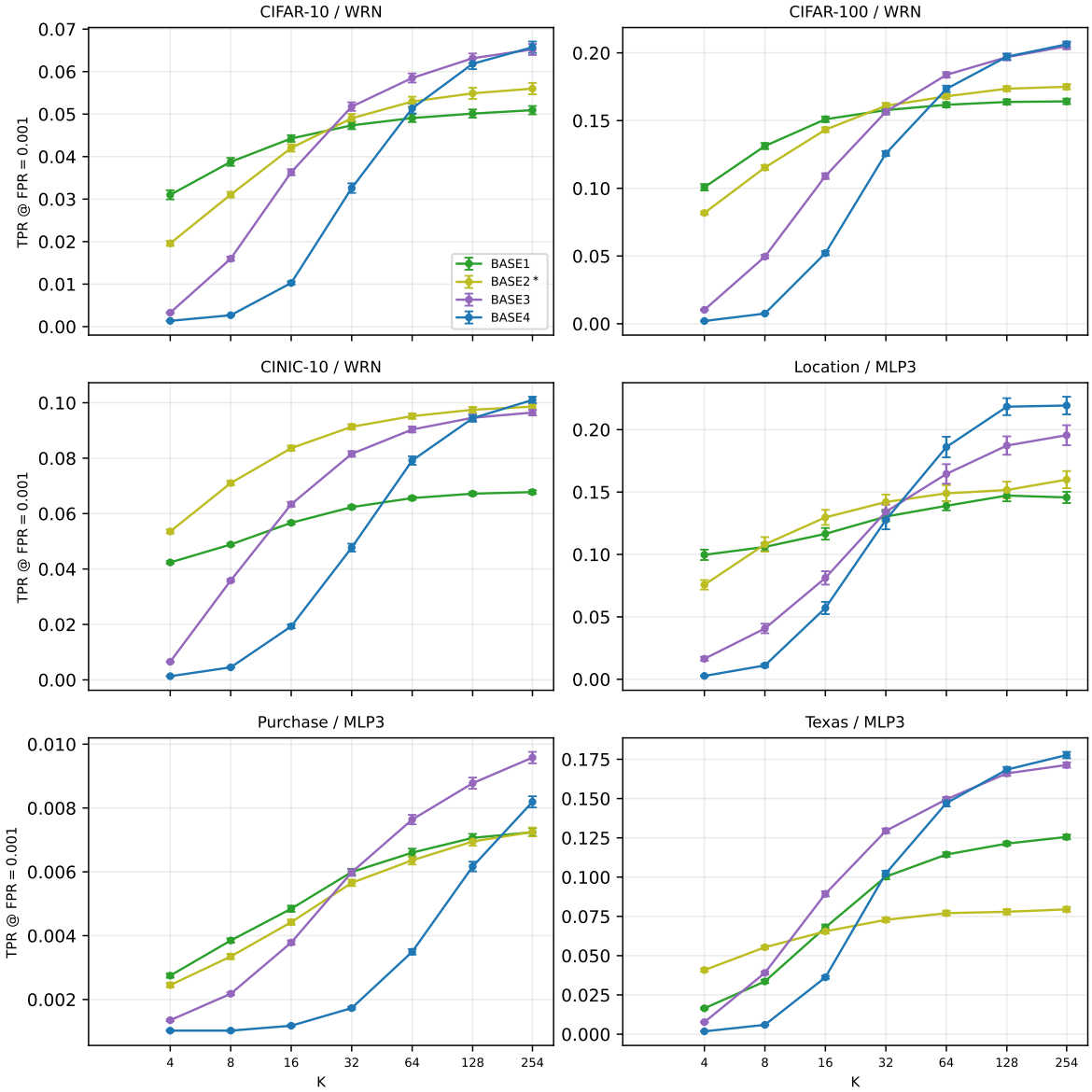

Figure 5: TPR at FPR= 0.001 vs. K, WRN / MLP3 testbeds. BASE1, BASE2*, BASE3, BASE4. Mean ± 1 SE over 32 replicates.

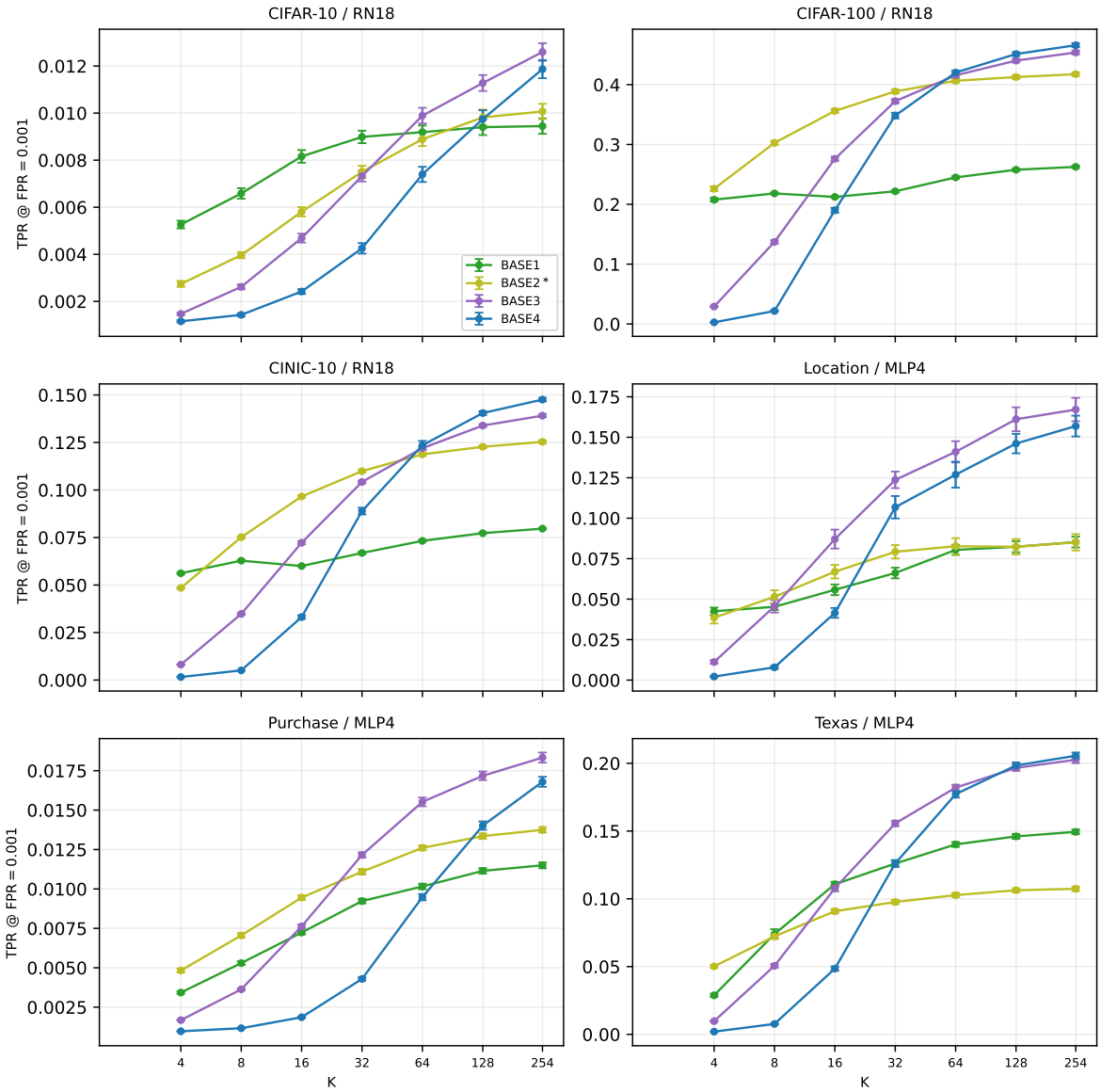

Figure 6: TPR at FPR= 0.001 vs. K, RN18 / MLP4 testbeds. BASE1, BASE2*, BASE3, BASE4. Mean ± 1 SE over 32 replicates.

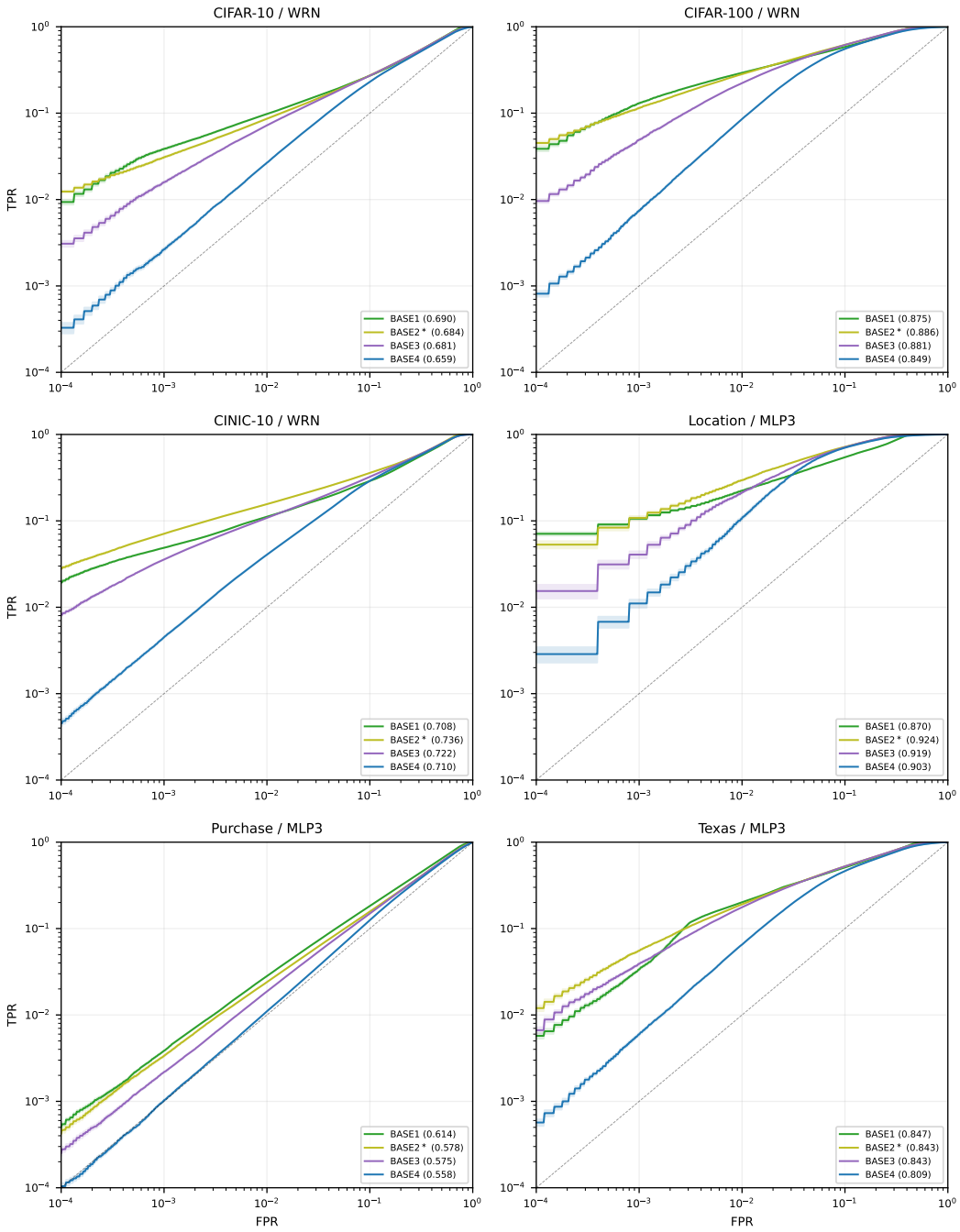

Figure 7: ROC at $K = 8$, WRN / MLP3 testbeds. All BASE methods overlaid (BASE1, BASE2*, BASE3, BASE4); mean ± 1 SE over 32 replicates. At small K the simpler families lead in the deep tail.

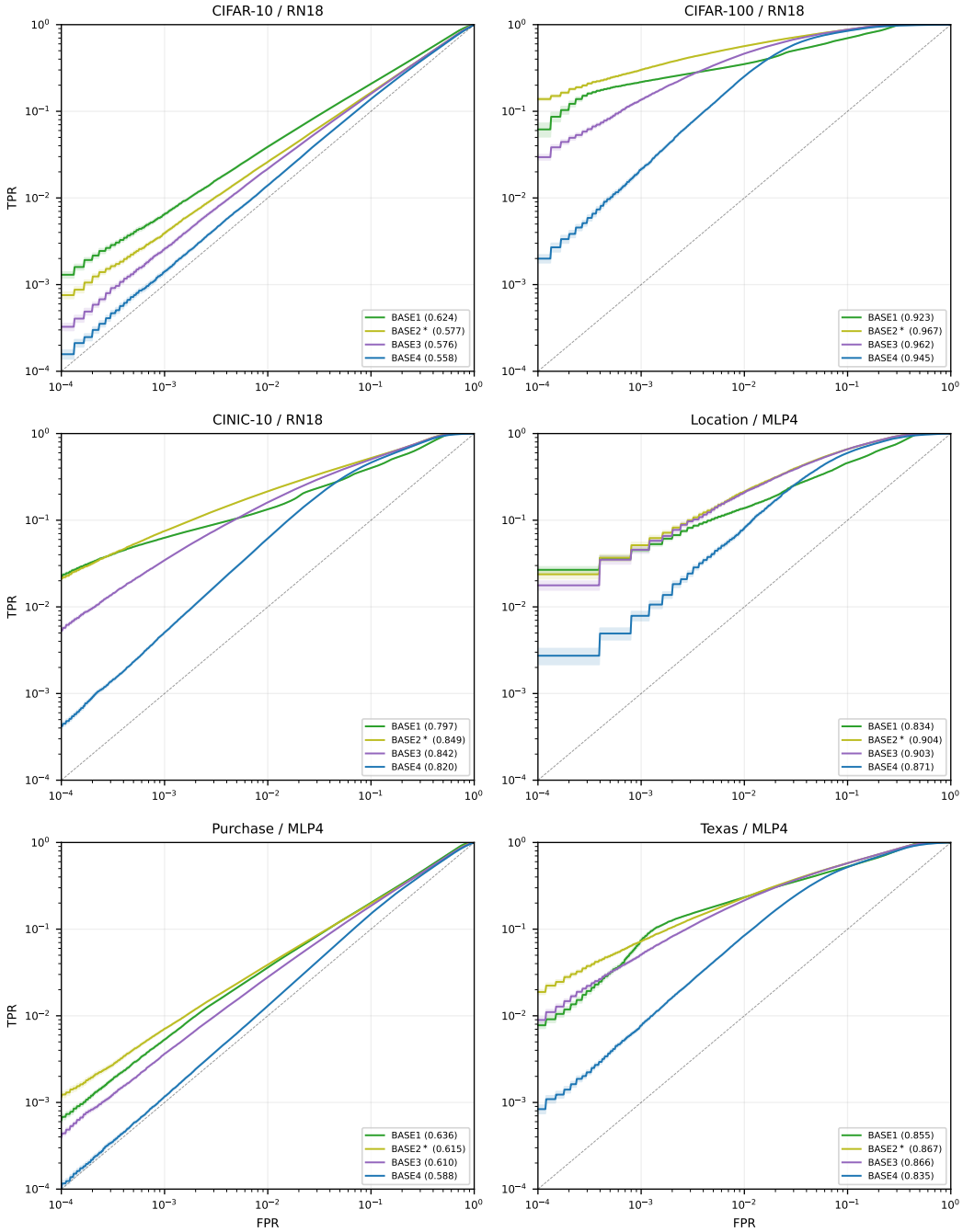

Figure 8: ROC at $K = 8$, RN18 / MLP4 testbeds. BASE1, BASE2*, BASE3, BASE4 overlaid; mean ± 1 SE over 32 replicates.

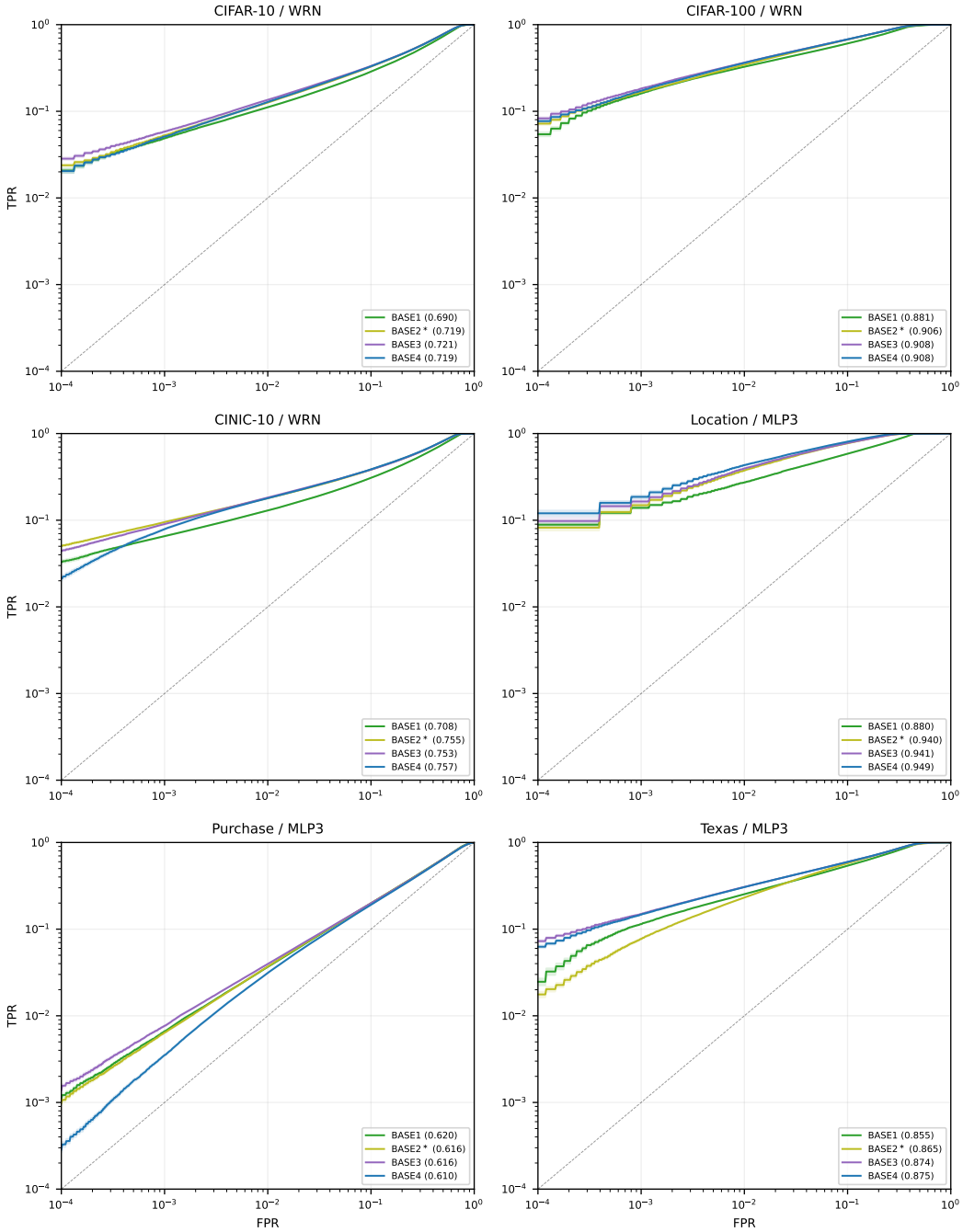

Figure 9: ROC at $K = 64$, WRN / MLP3 testbeds. BASE1, BASE2*, BASE3, BASE4 overlaid; mean ± 1 SE over 32 replicates.

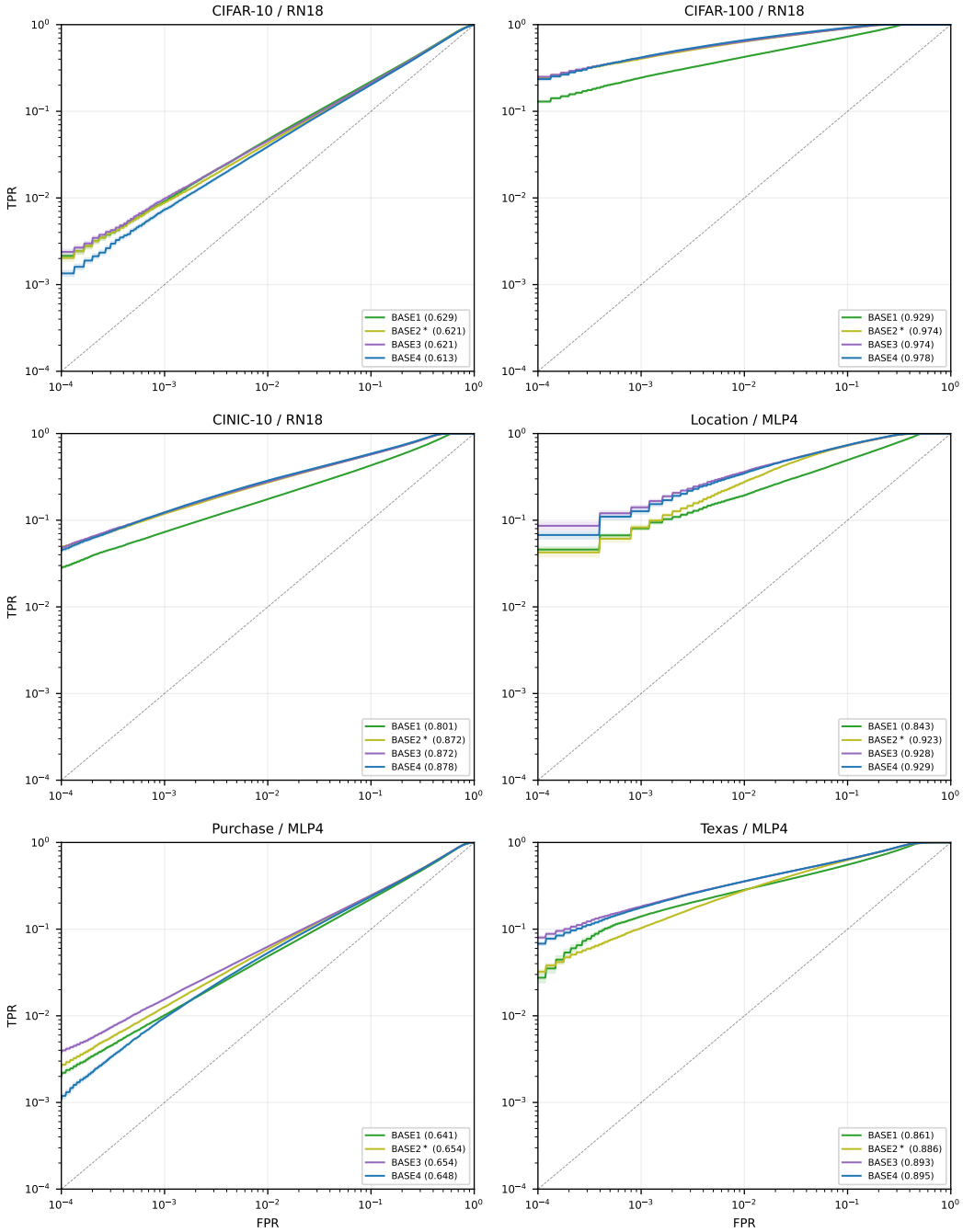

Figure 10: ROC at $K = 64$, RN18 / MLP4 testbeds. BASE1, BASE2*, BASE3, BASE4 overlaid; mean ± 1 SE over 32 replicates.

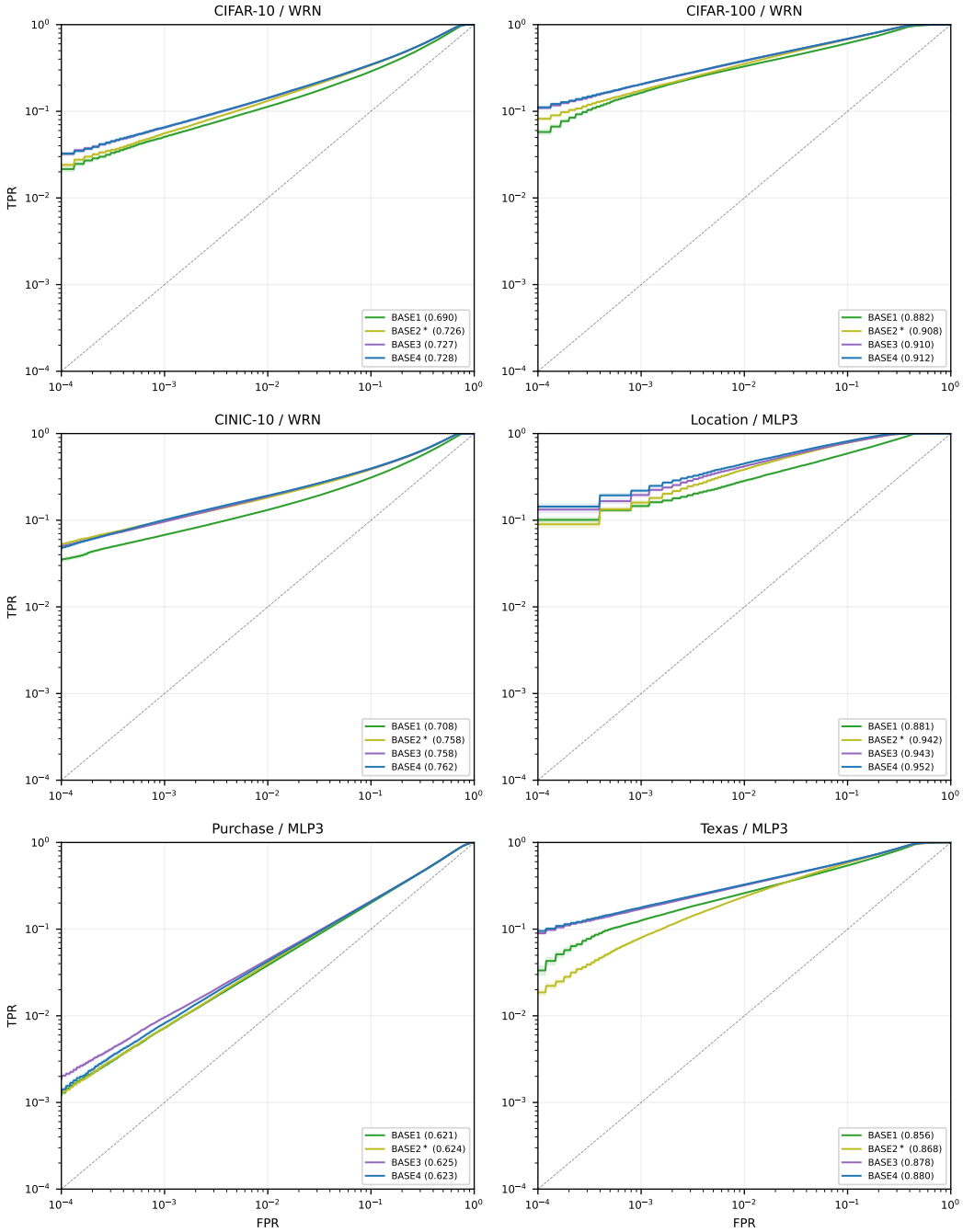

Figure 11: ROC at $K = 254$, WRN / MLP3 testbeds. BASE1, BASE2*, BASE3, BASE4 overlaid; mean ± 1 SE over 32 replicates. At large K the richer families lead, reversing the small-K ordering.

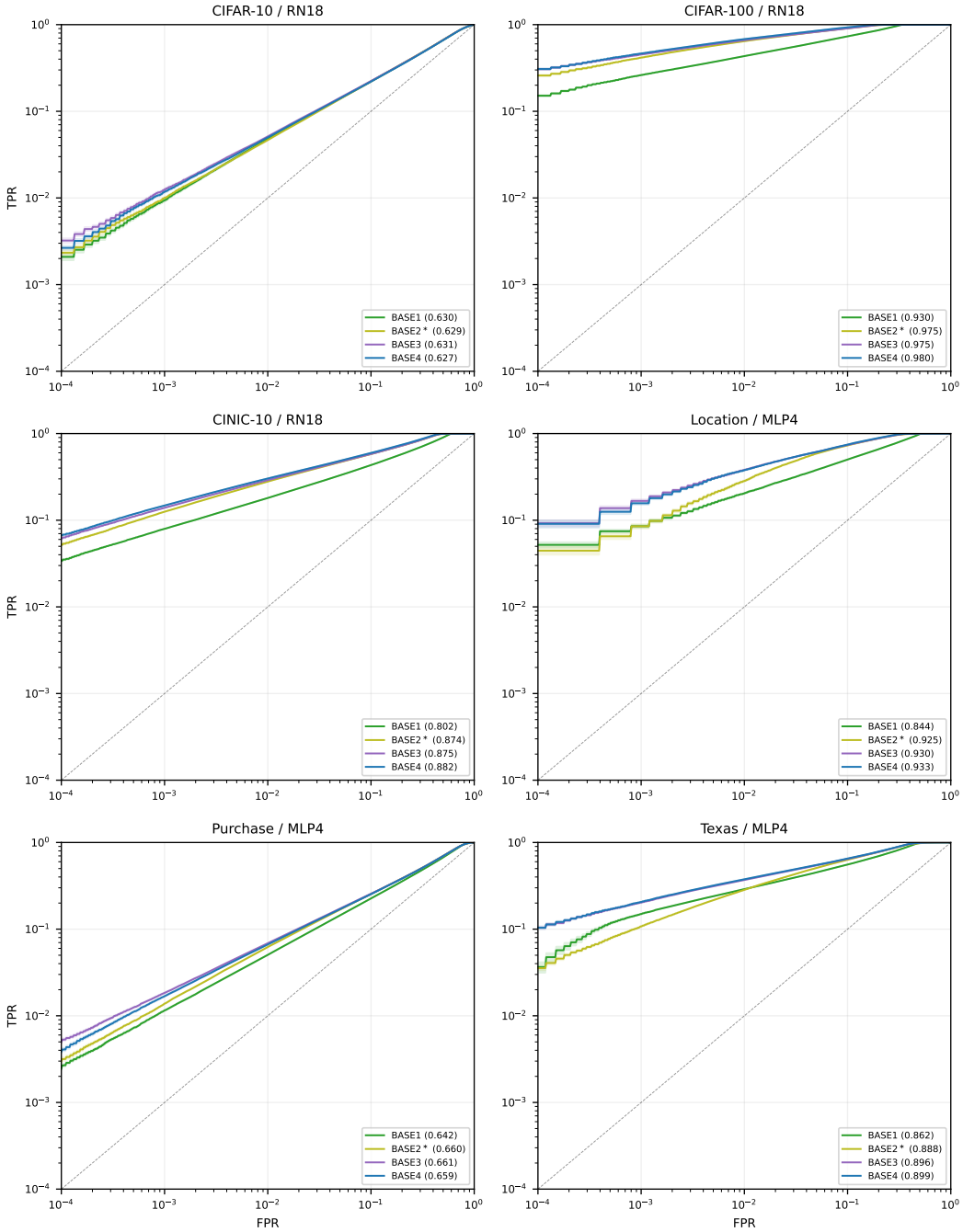

Figure 12: ROC at $K = 254$, RN18 / MLP4 testbeds. BASE1, BASE2*, BASE3, BASE4 overlaid; mean ± 1 SE over 32 replicates.

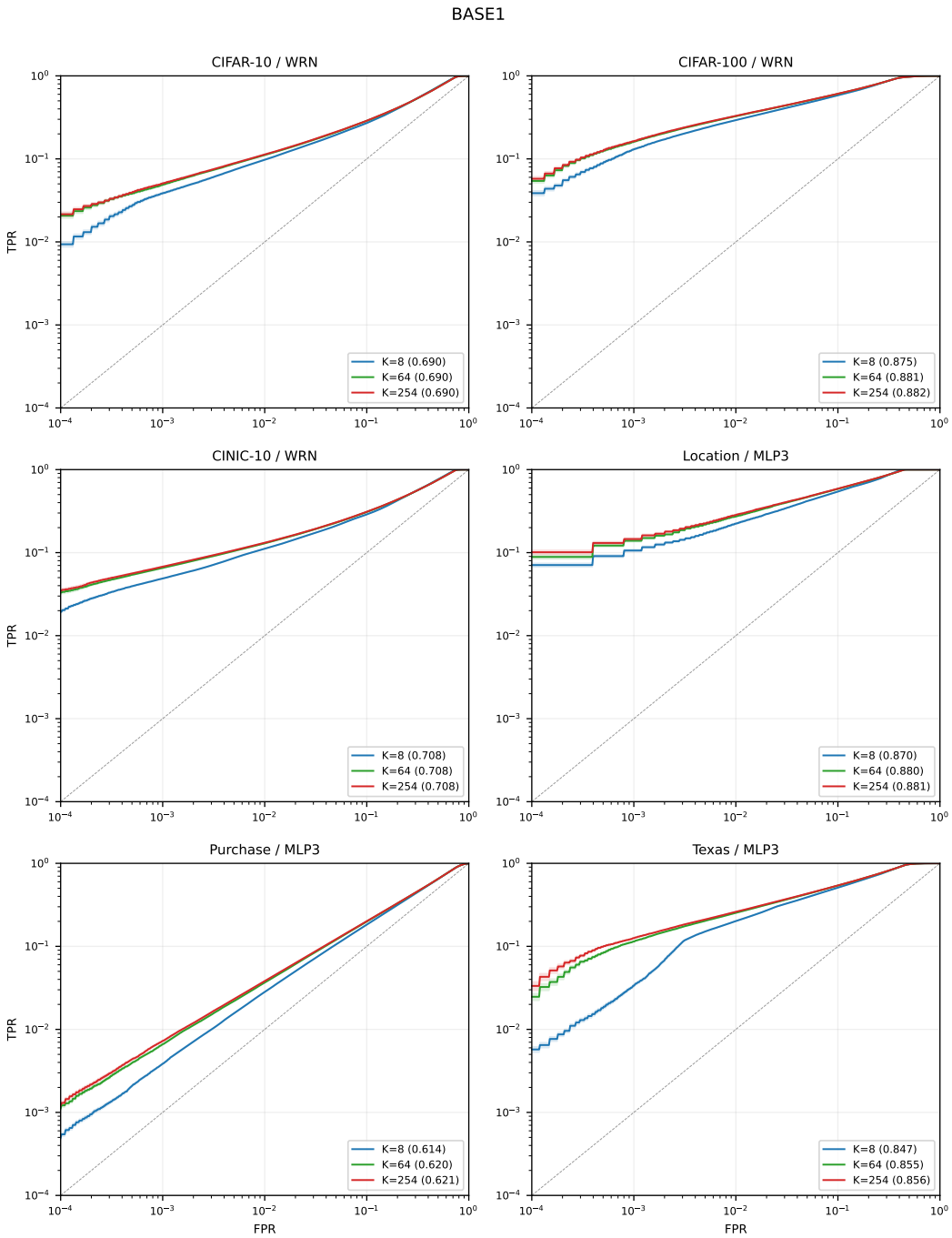

Figure 13: BASE1 ROC across $K \in \{4, 8, 16, 32, 64, 128, 254\}$, WRN / MLP3 testbeds. Mean ± 1 SE over 32 replicates.

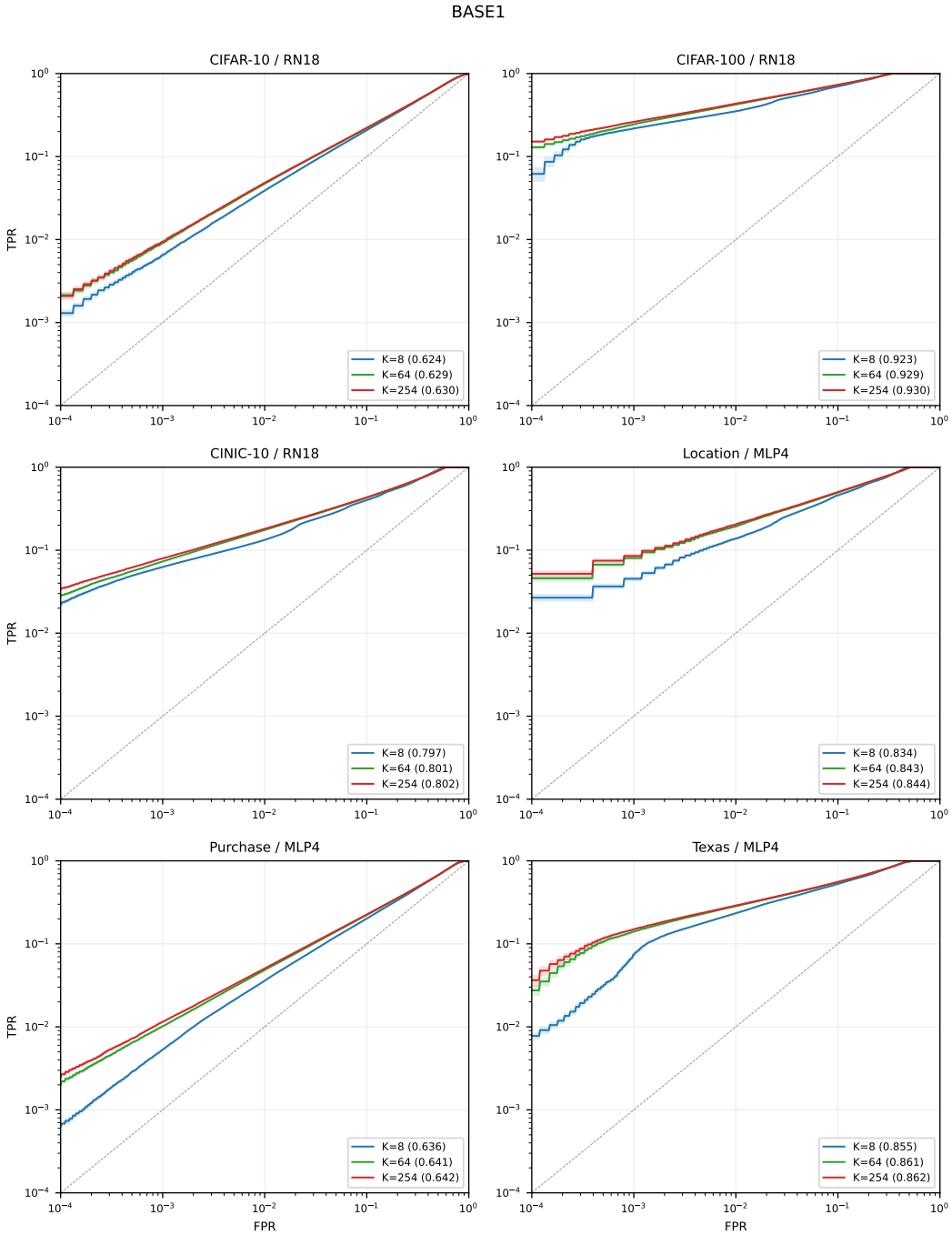

Figure 14: BASE1 ROC across K, RN18 / MLP4 testbeds. Mean ± 1 SE over 32 replicates.

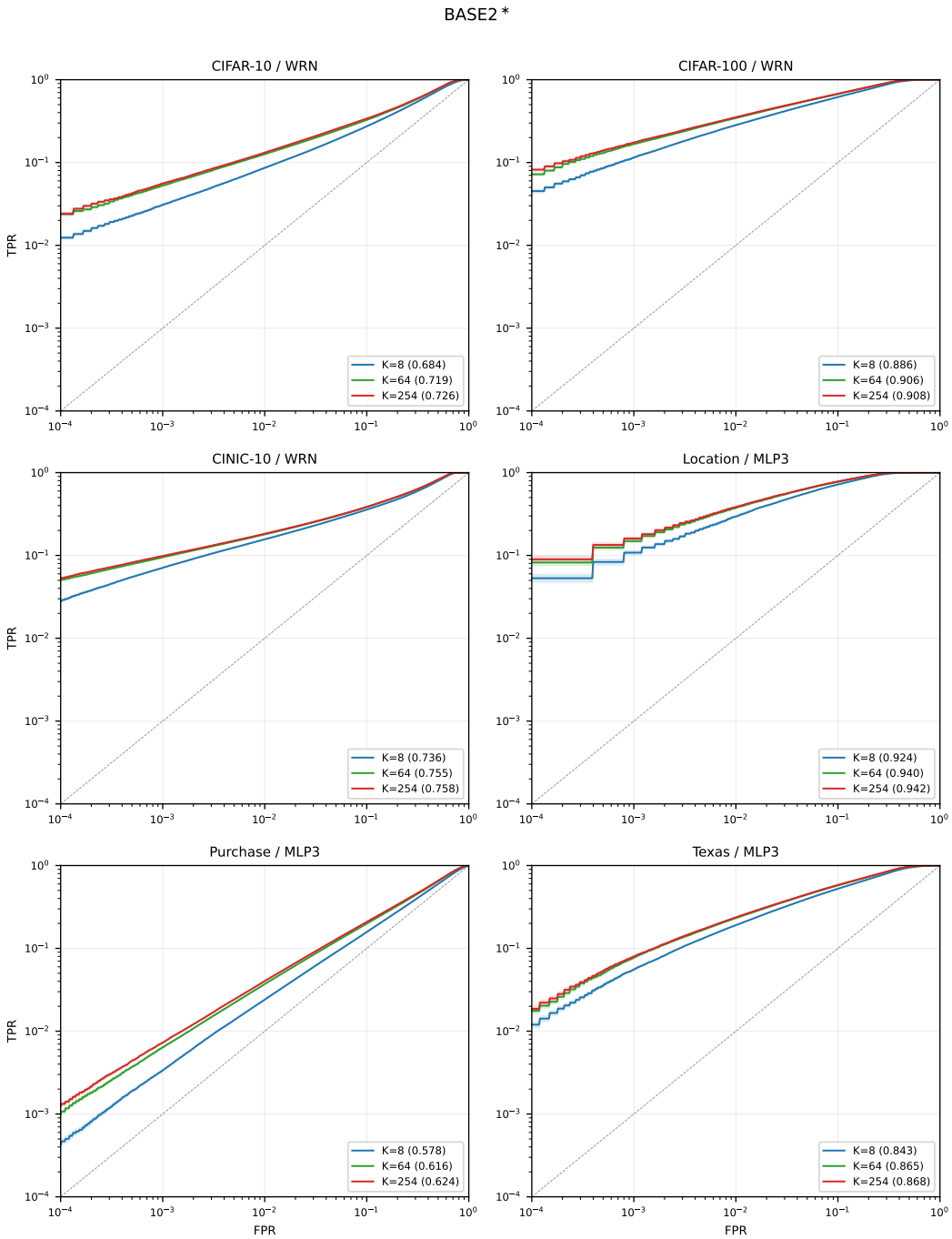

Figure 15: BASE2* ROC across K, WRN / MLP3 testbeds. Mean ± 1 SE over 32 replicates.

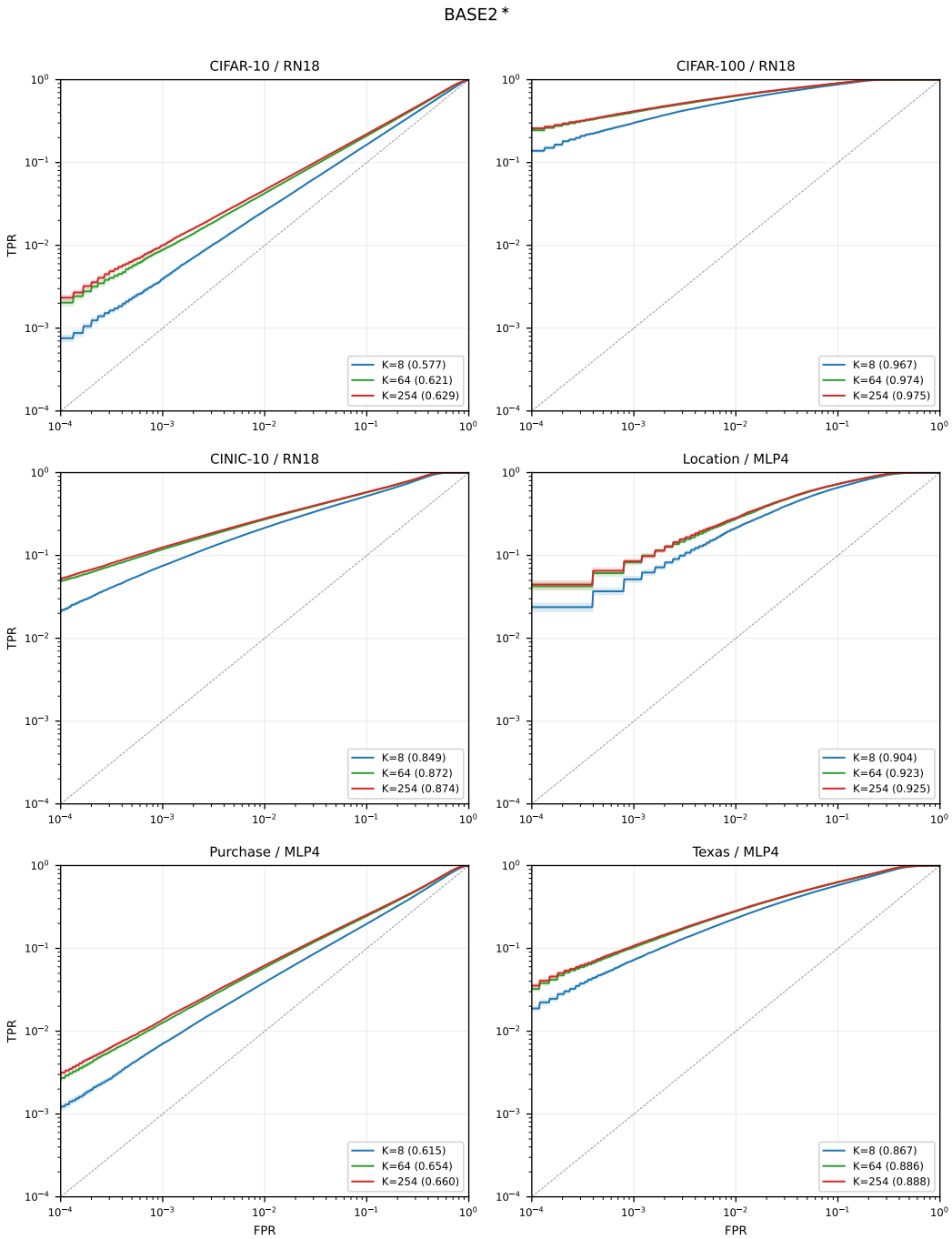

Figure 16: BASE2* ROC across K, RN18 / MLP4 testbeds. Mean ± 1 SE over 32 replicates.

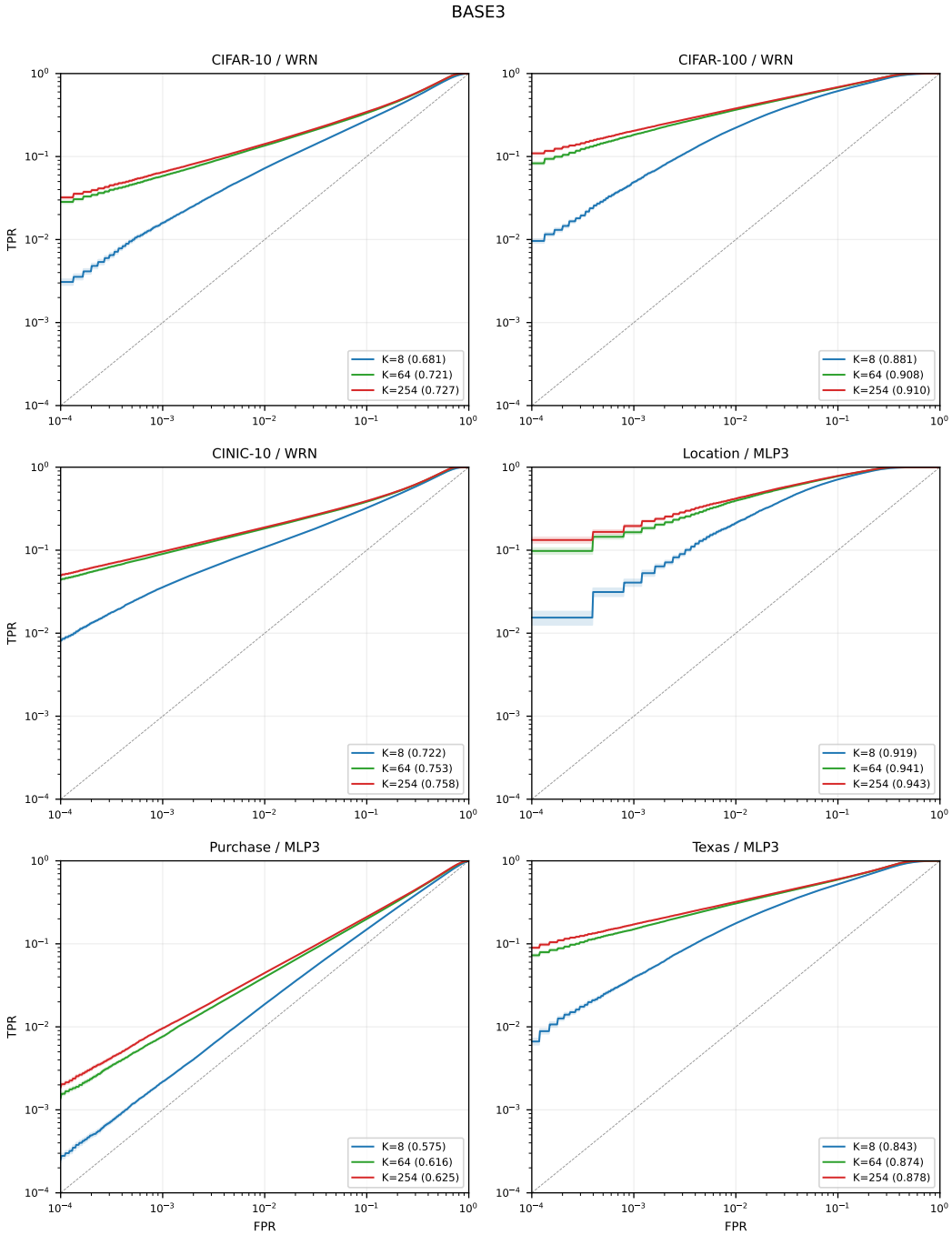

Figure 17: BASE3 ROC across K, WRN / MLP3 testbeds. Mean ± 1 SE over 32 replicates.

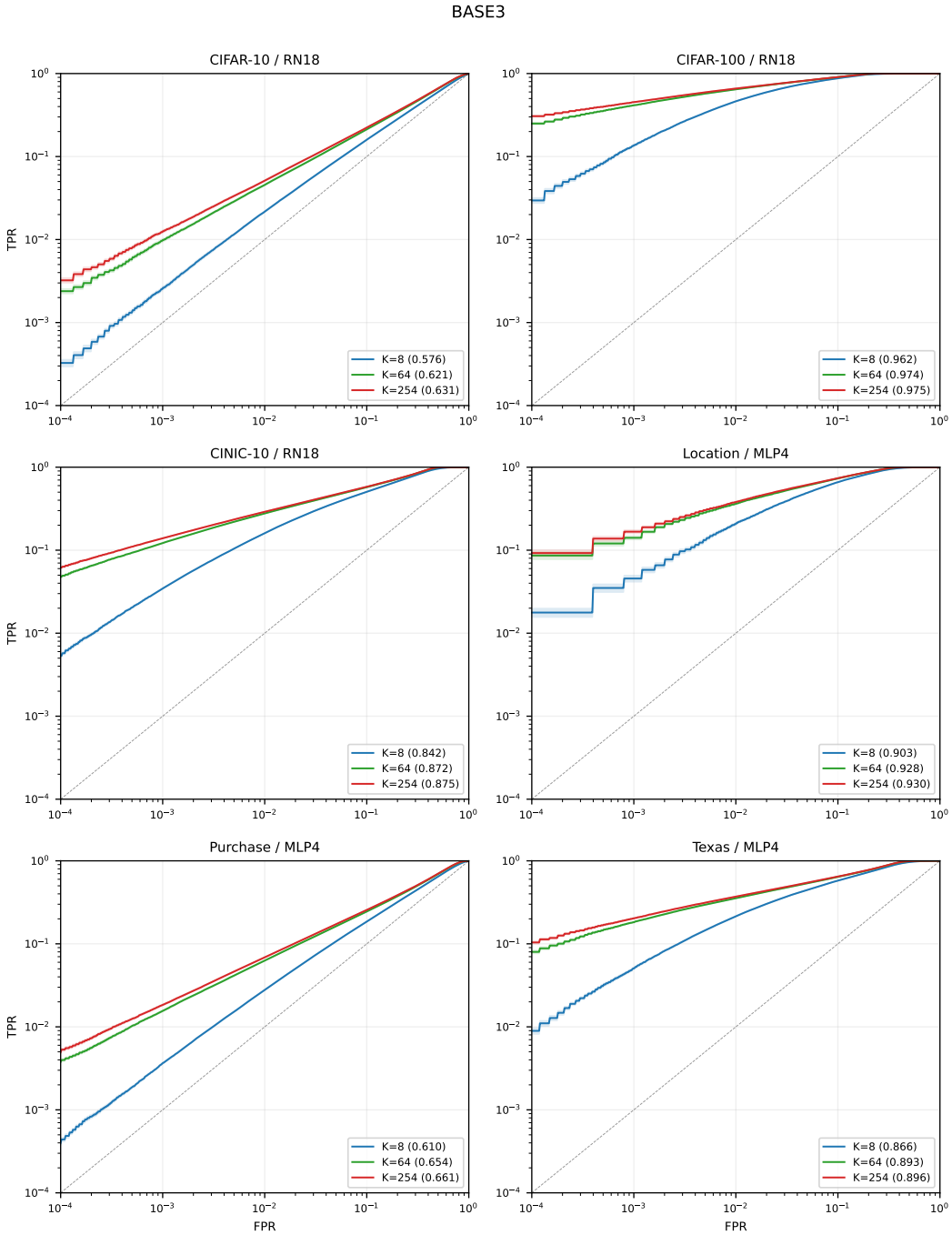

Figure 18: BASE3 ROC across K, RN18 / MLP4 testbeds. Mean ± 1 SE over 32 replicates.

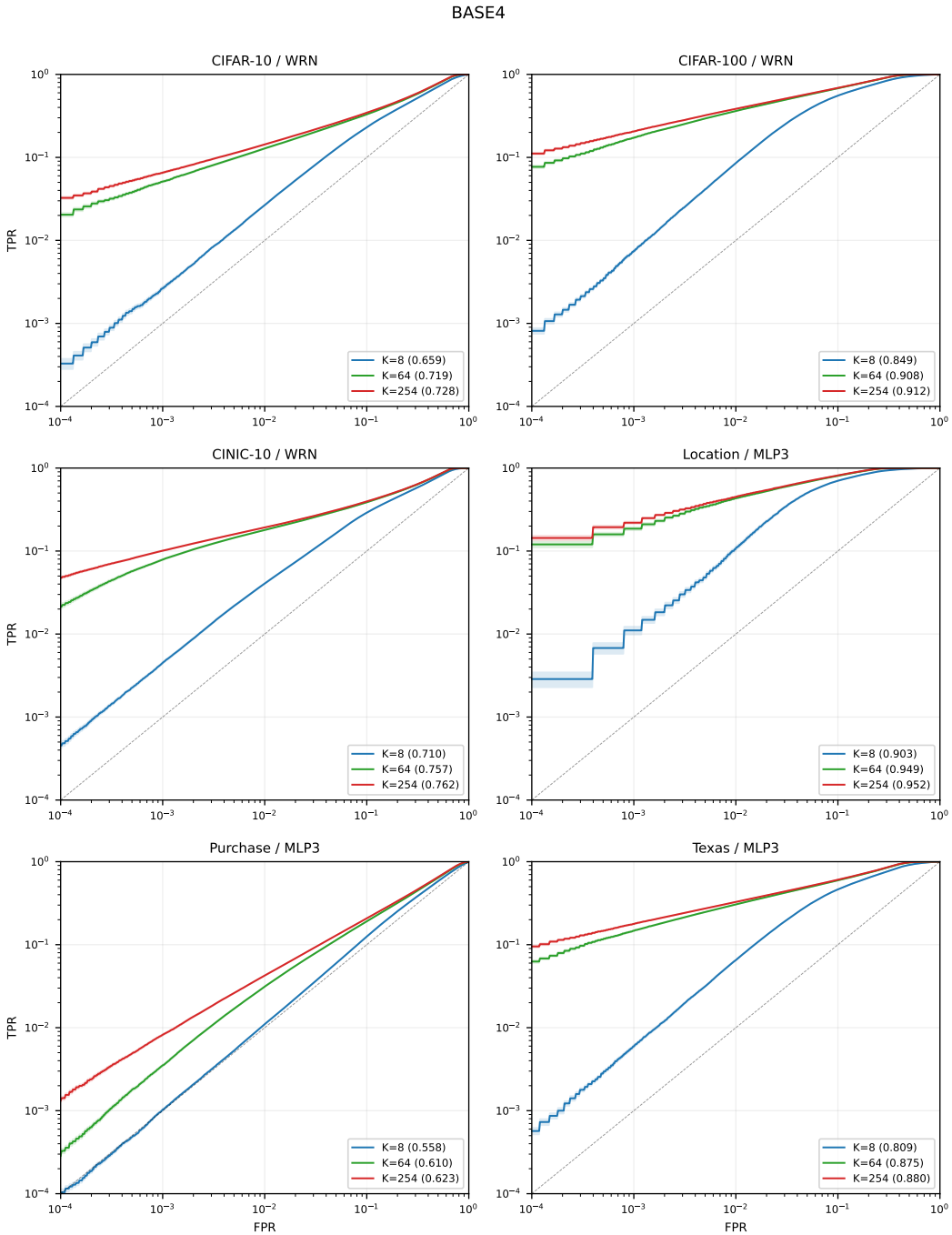

Figure 19: BASE4 ROC across K, WRN / MLP3 testbeds (BASE4 $\equiv$ LiRA online). Mean ± 1 SE over 32 replicates.

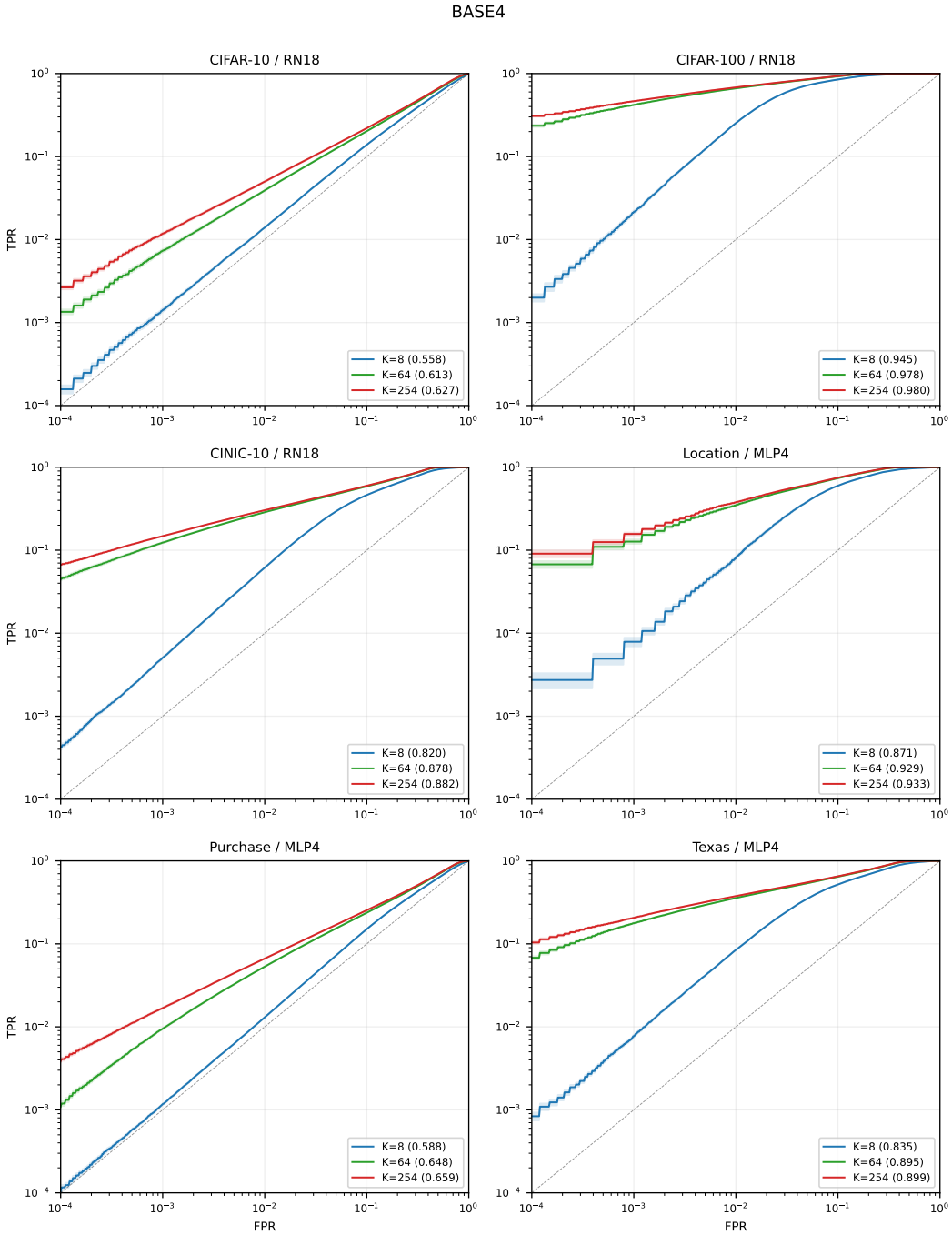

Figure 20: BASE4 ROC across K, RN18 / MLP4 testbeds. Mean ± 1 SE over 32 replicates.

Testbed	BASE1	BASE2*	BASE3	BASE4
CIFAR-10 / WRN	**0.690**	0.684	0.681	0.659
CIFAR-100 / WRN	0.875	**0.886**	0.881	0.849
CINIC-10 / WRN	0.708	**0.736**	0.722	0.710
Location / MLP3	0.870	**0.924**	0.919	0.903
Purchase / MLP3	**0.614**	0.578	0.575	0.558
Texas / MLP3	**0.847**	0.843	0.843	0.809
CIFAR-10 / RN18	**0.624**	0.577	0.576	0.558
CIFAR-100 / RN18	0.923	**0.967**	0.962	0.945
CINIC-10 / RN18	0.797	**0.849**	0.842	0.820
Location / MLP4	0.834	**0.904**	0.903	0.871
Purchase / MLP4	**0.636**	0.615	0.610	0.588
Texas / MLP4	0.855	**0.867**	0.866	0.835

Table 1: AUC, $K = 8$. Mean over 32 replicates. Best per row in bold.

Testbed	BASE1	BASE2*	BASE3	BASE4
CIFAR-10 / WRN	0.690	0.719	**0.721**	0.719
CIFAR-100 / WRN	0.881	0.906	0.908	**0.908**
CINIC-10 / WRN	0.708	0.755	0.753	**0.757**
Location / MLP3	0.880	0.940	0.941	**0.949**
Purchase / MLP3	**0.620**	0.616	0.616	0.610
Texas / MLP3	0.855	0.865	0.874	**0.875**
CIFAR-10 / RN18	**0.629**	0.621	0.621	0.613
CIFAR-100 / RN18	0.929	0.974	0.974	**0.978**
CINIC-10 / RN18	0.801	0.872	0.872	**0.878**
Location / MLP4	0.843	0.923	0.928	**0.929**
Purchase / MLP4	0.641	**0.654**	**0.654**	0.648
Texas / MLP4	0.861	0.886	0.893	**0.895**

Table 2: AUC, $K = 64$. Mean over 32 replicates. Best per row in bold.

Testbed	BASE1	BASE2*	BASE3	BASE4
CIFAR-10 / WRN	0.690	0.726	0.727	**0.728**
CIFAR-100 / WRN	0.882	0.908	0.910	**0.912**
CINIC-10 / WRN	0.708	0.758	0.758	**0.762**
Location / MLP3	0.881	0.942	0.943	**0.952**
Purchase / MLP3	0.621	0.624	**0.625**	0.623
Texas / MLP3	0.856	0.868	0.878	**0.880**
CIFAR-10 / RN18	0.630	0.629	**0.631**	0.627
CIFAR-100 / RN18	0.930	0.975	0.975	**0.980**
CINIC-10 / RN18	0.802	0.874	0.875	**0.882**
Location / MLP4	0.844	0.925	0.930	**0.933**
Purchase / MLP4	0.642	0.660	**0.661**	0.659
Texas / MLP4	0.862	0.888	0.896	**0.899**

Table 3: AUC, $K = 254$. Mean over 32 replicates. Best per row in bold.

Testbed	BASE1	BASE2*	BASE3	BASE4
CIFAR-10 / WRN	**0.098**	0.086	0.072	0.027
CIFAR-100 / WRN	**0.294**	0.283	0.223	0.086
CINIC-10 / WRN	0.111	**0.156**	0.109	0.041
Location / MLP3	0.225	**0.297**	0.215	0.110
Purchase / MLP3	**0.028**	0.024	0.019	0.011
Texas / MLP3	**0.202**	0.191	0.177	0.065
CIFAR-10 / RN18	**0.039**	0.026	0.022	0.014
CIFAR-100 / RN18	0.351	**0.566**	0.463	0.254
CINIC-10 / RN18	0.133	**0.215**	0.161	0.062
Location / MLP4	0.139	**0.217**	0.211	0.083
Purchase / MLP4	0.036	**0.039**	0.028	0.013
Texas / MLP4	**0.233**	0.231	0.216	0.084

Table 4: TPR @ FPR= 0.01, $K = 8$. Mean over 32 replicates. Best per row in bold.

Testbed	BASE1	BASE2*	BASE3	BASE4
CIFAR-10 / WRN	0.111	0.126	**0.136**	0.128
CIFAR-100 / WRN	0.326	0.345	**0.366**	0.362
CINIC-10 / WRN	0.129	0.180	**0.182**	0.179
Location / MLP3	0.274	0.378	0.398	**0.434**
Purchase / MLP3	0.037	0.037	**0.040**	0.031
Texas / MLP3	0.253	0.232	**0.306**	0.305
CIFAR-10 / RN18	**0.047**	0.043	0.046	0.039
CIFAR-100 / RN18	0.425	0.636	0.645	**0.664**
CINIC-10 / RN18	0.175	0.272	0.277	**0.287**
Location / MLP4	0.195	0.280	**0.364**	0.351
Purchase / MLP4	0.048	0.059	**0.063**	0.053
Texas / MLP4	0.284	0.279	0.355	**0.357**

Table 5: TPR @ FPR= 0.01, $K = 64$. Mean over 32 replicates. Best per row in bold.

Testbed	BASE1	BASE2*	BASE3	BASE4
CIFAR-10 / WRN	0.113	0.132	0.142	**0.143**
CIFAR-100 / WRN	0.330	0.354	0.383	**0.385**
CINIC-10 / WRN	0.131	0.183	0.189	**0.192**
Location / MLP3	0.286	0.387	0.421	**0.452**
Purchase / MLP3	0.038	0.040	**0.045**	0.043
Texas / MLP3	0.261	0.237	0.320	**0.326**
CIFAR-10 / RN18	0.048	0.047	**0.051**	0.050
CIFAR-100 / RN18	0.434	0.645	0.660	**0.682**
CINIC-10 / RN18	0.181	0.280	0.290	**0.304**
Location / MLP4	0.206	0.286	**0.383**	0.381
Purchase / MLP4	0.050	0.062	**0.069**	0.066
Texas / MLP4	0.290	0.284	0.371	**0.376**

Table 6: TPR @ FPR= 0.01, $K = 254$. Mean over 32 replicates. Best per row in bold.

Testbed	BASE1	BASE2*	BASE3	BASE4
CIFAR-10 / WRN	**0.039**	0.031	0.016	0.003
CIFAR-100 / WRN	**0.131**	0.115	0.050	0.008
CINIC-10 / WRN	0.049	**0.071**	0.036	0.005
Location / MLP3	0.106	**0.108**	0.041	0.011
Purchase / MLP3	**0.004**	0.003	0.002	0.001
Texas / MLP3	0.034	**0.055**	0.039	0.006
CIFAR-10 / RN18	**0.007**	0.004	0.003	0.001
CIFAR-100 / RN18	0.218	**0.303**	0.137	0.022
CINIC-10 / RN18	0.063	**0.075**	0.035	0.005
Location / MLP4	0.045	**0.051**	0.046	0.008
Purchase / MLP4	0.005	**0.007**	0.004	0.001
Texas / MLP4	**0.074**	0.072	0.051	0.008

Table 7: TPR @ FPR= 0.001, $K = 8$. Mean over 32 replicates. Best per row in bold.

Testbed	BASE1	BASE2*	BASE3	BASE4
CIFAR-10 / WRN	0.049	0.053	**0.059**	0.051
CIFAR-100 / WRN	0.162	0.168	**0.184**	0.173
CINIC-10 / WRN	0.066	**0.095**	0.090	0.079
Location / MLP3	0.139	0.149	0.164	**0.186**
Purchase / MLP3	0.007	0.006	**0.008**	0.003
Texas / MLP3	0.114	0.077	**0.150**	0.147
CIFAR-10 / RN18	0.009	0.009	**0.010**	0.007
CIFAR-100 / RN18	0.245	0.406	0.416	**0.420**
CINIC-10 / RN18	0.073	0.119	0.122	**0.124**
Location / MLP4	0.080	0.083	**0.141**	0.127
Purchase / MLP4	0.010	0.013	**0.016**	0.009
Texas / MLP4	0.140	0.103	**0.182**	0.177

Table 8: TPR @ FPR= 0.001, $K = 64$. Mean over 32 replicates. Best per row in bold.

Testbed	BASE1	BASE2*	BASE3	BASE4
CIFAR-10 / WRN	0.051	0.056	0.065	**0.066**
CIFAR-100 / WRN	0.164	0.175	0.205	**0.206**
CINIC-10 / WRN	0.068	0.099	0.096	**0.101**
Location / MLP3	0.146	0.160	0.196	**0.219**
Purchase / MLP3	0.007	0.007	**0.010**	0.008
Texas / MLP3	0.126	0.079	0.171	**0.178**
CIFAR-10 / RN18	0.009	0.010	**0.013**	0.012
CIFAR-100 / RN18	0.263	0.417	0.454	**0.466**
CINIC-10 / RN18	0.080	0.125	0.139	**0.148**
Location / MLP4	0.085	0.085	**0.167**	0.157
Purchase / MLP4	0.011	0.014	**0.018**	0.017
Texas / MLP4	0.149	0.107	0.203	**0.206**

Table 9: TPR @ FPR= 0.001, $K = 254$. Mean over 32 replicates. Best per row in bold.